

Omprakash: Alviso Energy Audit

Snigdha Gurram

Department of Engineering, Shiley-Marcos School of Engineering

EHS 520: Sustainable Energy

Prof. Jon Colon

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In the 2022-2023 fiscal year report done by the California State Library, made a Building Forward Library Facilities Improvement Program which is a grant program for libraries in California. One library that is a part of this program is Alviso Branch Library which received \$614,374 and has been used for updates to the security, HVAC, and lighting systems along with providing interior upgrades mainly towards accessible access (California State Library, 2024). The estimated completion date is March 31, 2026 (ibid.).

Energy Data Collection:

15.4 kWh/sq ft / year baseline from (EIA, 2022)

Hourly consumption for a full week: $4,460 \text{ sq ft} \times 15.4 \text{ kWh/sq ft/year} = 68,684 \text{ kWh/year} / 52 \text{ weeks} = \text{about } 1,321.24 \text{ kWh/week} / 7 \text{ days} = 188.7 \text{ kWh/day}$

Alviso Branch Library Hours:

Monday: Operating hours: 4 hours; Off hours: 20 hours

2:00PM - 6:00PM

Tuesday: Operating hours: 9 hours; Off hours: 15 hours

10:00AM - 7:00PM

Wednesday: Operating hours: 9 hours; Off hours: 15 hours

10:00AM - 7:00PM

Thursday: Operating hours: 9 hours; Off hours: 15 hours

10:00AM - 7:00PM

Friday: Operating hours: 4 hours; Off hours: 20 hours

2:00PM - 6:00PM

Saturday: Operating hours: 8 hours; Off hours: 16 hours

10:00AM - 6:00PM

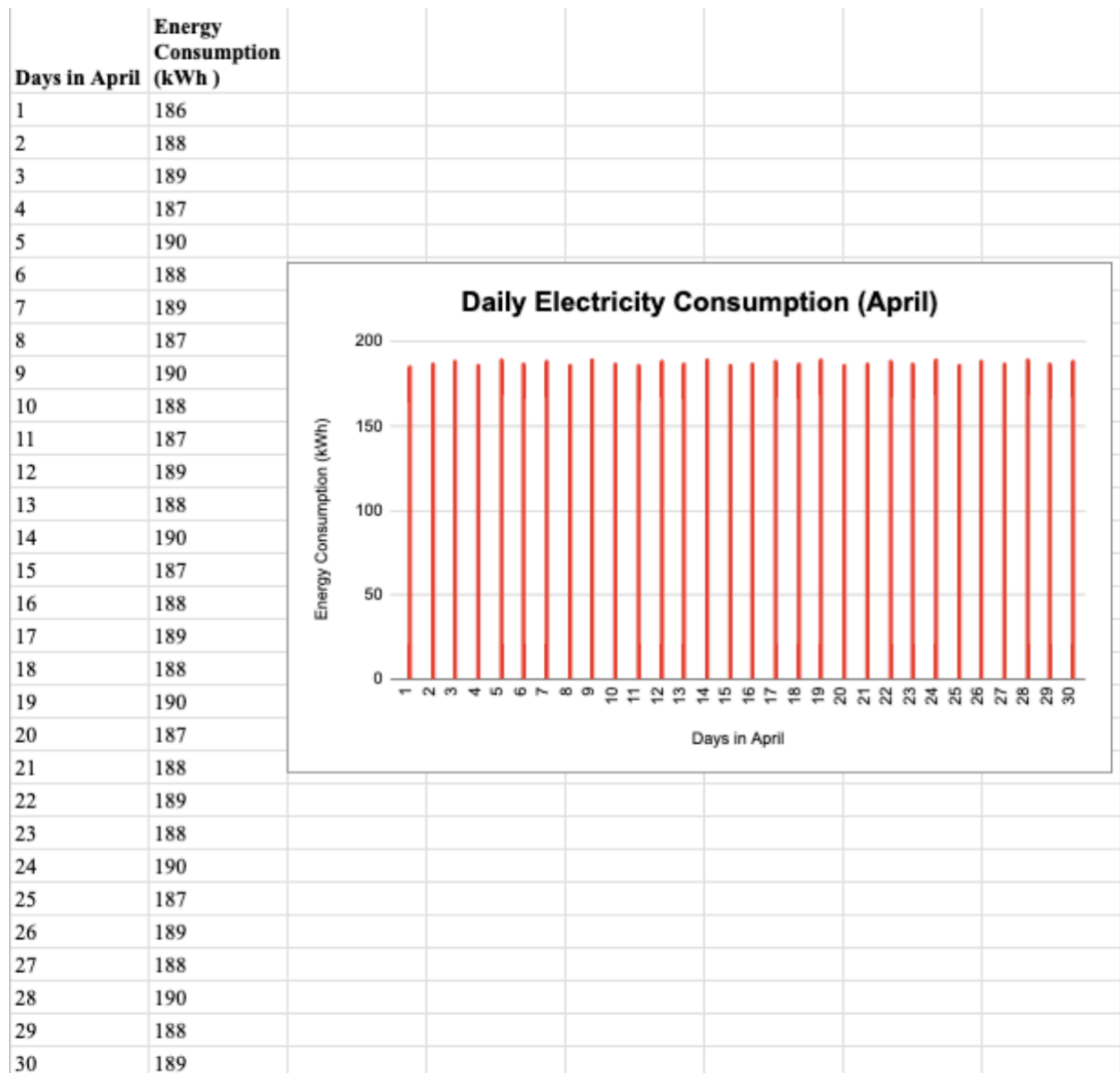
Sunday: Operating hours: 7 hours; Off hours: 17 hours

12:00PM - 7:00PM

Days	Operating Hours	Off Hours	Operating Energy	Off Hours Energy	Daily Total
Monday	4 hours	20 hours	59.16 kWh	98.6 kWh	157.76 kWh
Tuesday	9 hours	15 hours	133.11 kWh	73.95 kWh	207.06 kWh
Wednesday	9 hours	15 hours	133.11 kWh	73.95 kWh	207.06 kWh
Thursday	9 hours	15 hours	133.11 kWh	73.95 kWh	207.06 kWh
Friday	4 hours	20 hours	59.16 kWh	98.6 kWh	157.76 kWh
Saturday	8 hours	16 hours	118.32 kWh	78.88 kWh	197.20 kWh
Sunday	7 hours	17 hours	103.53 kWh	83.81 kWh	187.34 kWh
Totals	50 hours	118 hours	739.5 kWh	581.74 kWh	1,321.24 kWh/week

Hourly consumption for a full week

Average temperature for April is around 65 degrees Fahrenheit for San Jose California with the weekly baseline being 188.7 kWh/day.



Daily consumption over a month

Months (3 months only)	Monthly Total
April (Spring)	5,640 kWh
June (Early Summer)	6,240 kWh
July (Peak Summer)	6,913 kWh

Determine annual consumption based on the data:
 $4,460 \text{ sq ft} \times 15.4 \text{ kWh/sq ft/year} = 68,684 \text{ kWh/year}$
which rounds to about 70,000 kWh per year

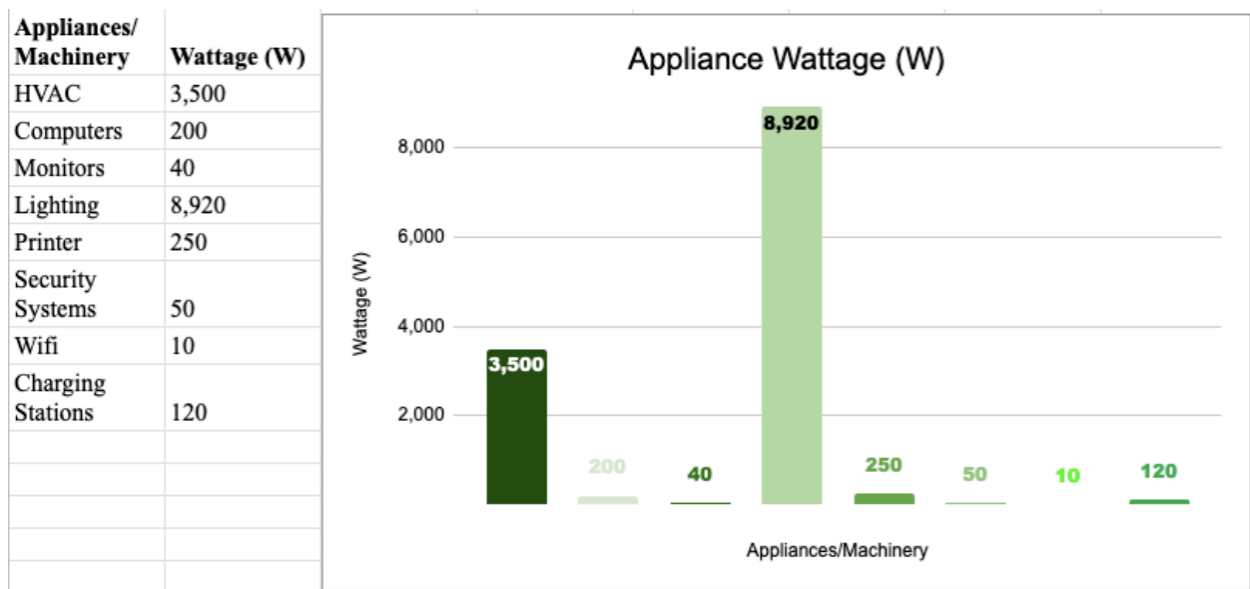
Energy Analysis and Calculations:

The times of peak power usage would be Monday from 2:00PM - 6:00PM, Tuesday from 10:00AM - 7:00PM, Wednesday from 10:00AM - 7:00PM, Thursday from 10:00AM - 7:00PM, Friday from 2:00PM - 6:00PM, Saturday from 10:00AM - 6:00PM, and Sunday from 12:00PM - 7:00PM. These times are correlated to the highest electricity demands that are between 14 - 15 kWh per hour while the off hours are around 5 kWh. This would very much likely increase during the summer months because of the cooling systems. The peak consumption happens during these times because of activities like computer and printers running, study rooms being used, and charging of devices increasing. The lighting systems that would be needed for reading areas, hallways, work areas, and bookshelves which is about 25% of the electricity usage in libraries. Heating, ventilation, and air conditioning are also working in the library in order to maintain the indoor temperature, circulate fresh air, and cool the library during the summer months. The HVAC would be in even more demand during the afternoon when the temperature peaks which causes the air conditioning and cooling systems to be in more demand. The HVAC is usually the largest consumer of energy in buildings.

The difference in daily consumption between weekdays and weekends is because of the changes in the hours and activities that happen. From Tuesday to Thursday, the highest daily energy consumption would occur with about 207 kWh per day because of the library being open for 9 hours. During this time lighting, computers, printers, and HVAC systems will all be operating to support the people coming to the library and the staff working there. With the increase in people means more devices being used and charged which increases the demand for electricity. Monday and Friday however are the lowest energy consumption out of the whole week with about 158 kWh per day since the library only opens for 4 hours. Even though the energy usage is a lot more during Tuesday to Thursday, the average for the weekdays are lowered a little bit. On the weekends the energy consumption is varied with Saturday using about 197 kWh per day since it is open for 8 hours and Sunday using about 187 kWh per day since it is open for 7 hours. As a result, without other activities, the weekday and weekend consumption somewhat comes to a similar energy consumption. The main source of energy consumption difference is because of the operating hours and the number of people at the library. Additionally, the weekdays would most likely be filled with people studying, researching, or working in some way which results in the lighting, computers, and HVAC running more for both the increase in people and the library hours. While the weekend has less hours and people would most like to travel outside of the town resulting in lower number of people and electricity consumption.

Energy Usage Audit:

Appliances/Machinery	Wattage (W)	Number of Hours per Week
HVAC	3,500	50
Computers	200	50
Monitors	40	50
Lighting	8,920	50
Printer	250	15
Security Systems	50	168
Wifi	10	168
Charging Stations	120	40



The main electricity consuming equipment are lighting, HVAC, computers, printers, charging stations, and networking systems. Lighting is one of the largest contributors to energy consumption with libraries needing about 1 - 3 watts per square foot of lighting power (HVAC, 2025). HVAC systems are large contributors especially during the warm seasons to cool the indoor area.

The difference in usage from the weekdays and the weekends are that Tuesday to Thursday, for 9 hours, and Saturday, 8 hours, and Sunday, 7 hours, are on the higher end for the number of hours the library is open which means the appliances/machinery are operating for longer. These days contribute to a total of about 70% of the number of hours they are operating for during the week.

While Monday and Friday operate for 4 hours resulting in less than 30% of the hours that they are being used during the week contributing from these days.

Per Capita Energy Estimate:

Assuming there is an average of 50 visitors per week:

$50 \text{ visitors} \times 52 \text{ weeks} = 2,600 \text{ visitors/year}$

$68,684 \text{ kWh/year} / 2,600 \text{ visitors/year} = 26.41 \text{ kWh per visitor}$

Energy Efficiency:

A plan to reduce the energy consumption by a minimum of 25% would be to have sensor lighting in the library. Since the library is an older build and is currently going through updates it would be the perfect time to change the lighting all throughout to sensor based lighting. This can help reduce the electricity consumption when there is no one in the area. Another sensor that can also be added to the lighting in daylight sensors so they turn on when it is a little dark or dull outside. This way natural light can be used as the lighting instead so some electricity can be saved. Another possibility is installing solar panels to the rooftop of the library which can help offset quite a bit of the electricity consumption. As this is an older building, improving the insulation can help to reduce the demand on the HVAC which in turn helps to reduce the electricity consumption.

The Alviso Branch Library has an important role in the community and is a part of the town's history. This audit shows that the HVAC systems, lighting, and equipment are the main source of electricity consumption. By adding in energy efficient upgrades and renewable energy sources the library can lower energy use and help to create more sustainable and resilient communities that can continue to use the long standing library.

References:

Alviso. San Jose Public Library. (n.d.). <https://www.sjpl.org/locations/alviso/>

California State Library. (2024, April). The California State Library's Building Forward Library. <https://www.library.ca.gov/uploads/2024/04/22-23-Building-Forward-Report.pdf>

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Lighting rules of thumb (HVAC) - HVAC-R & solar engineering resource. HVAC. (2025, May 16). <https://hvac-eng.com/lighting-rules-of-thumb/>